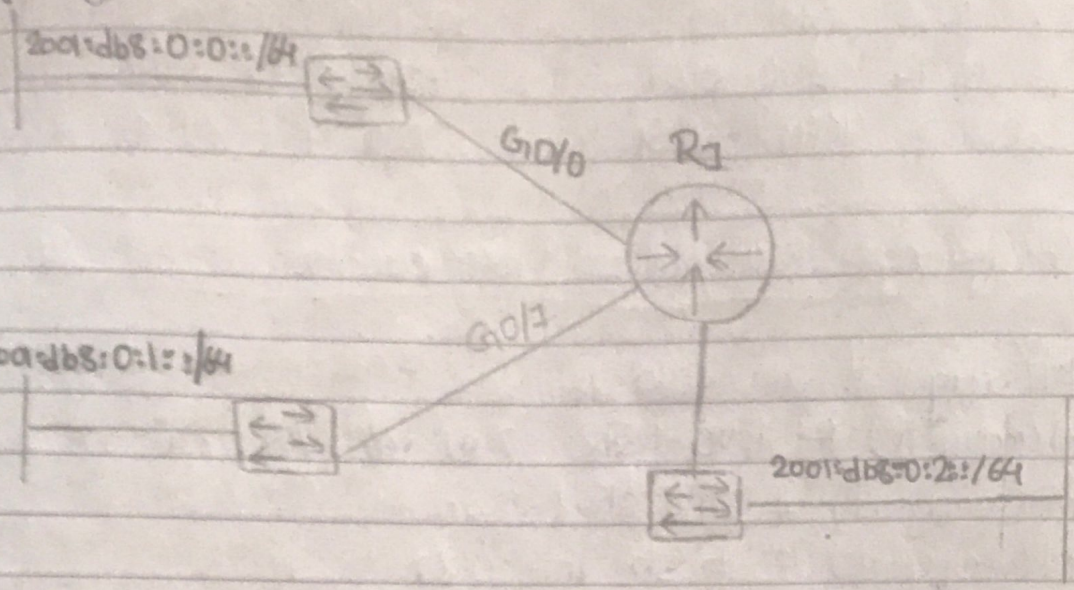
Day-32 27th January

~~CCNA~~ IPV6 Part-2

Configuring IPV6 address (EUI-64).



- EUI stands for Extended Unique Identifier.
- (Modified) EUI-64 is a method of converting MAC address (48 bits) to a 64 bit interface identifier.

- This interface identifier can then become the 'host portion' of a /64 IPv6 address.

- How to convert MAC add to a EUI-64 interface identifier.
  1) Divide the MAC address in half. Eg: MAC: 1234 5678 90AB
     Dividing 1234 56 | 78 90AB

  2) Insert FFFE in the middle.
     1234 56 FF FE 78 90AB.

  3) Invert the 7th bit, if 7th bit is 0, make it one & vice versa
     1234 56FF FE78 90AB → 1034 56FF FE78 90AB
     2 → 0010 → 7th bit=1 After convertion 7th bit → 0000

So, our int. id = **1034 56FF FE78 90AB.**

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Some examples

MAC address	EUI-64 interface identifier
782B CBAC 0867	7A2B CBFF FEAC 0867
0200 4C4F 4F50	0000 4CFF 6FE4F 4F50

Configuring IPv6 address using eui-64

- R1(config) # int g0/0
- R1(config-if) # ipv6 address 2001:db8::/64 eui-64
- R1(config-if) # no shutdown.

Checking Part

R1(config-if) # do sh int g0/0.
(From here see MAC)
Then, see IPv6 address to know eui-part with command.
R1(config-if) # do sh ipv6 int brief.

Why invert 7th bit? (Not necessary for CCNA).

- MAC addresses can be divided into two types:
  - ↳ UAA (Universally Administered Address).
    - ↳ Uniquely assigned to device by manufacturer.
  - ↳ LAA (Locally Administered Address).
    - ↳ Manually assigned by an admin (with the mac-address command on the interface). Doesn't have to be globally unique.

Note:

* In Cisco IOS we can manually configure mac using MAC-ADDRESS command.

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- We can identify a UAA or LAA by 7th bit of MAC address, called the U/L bit (Universal/Local bit):
  - ↳ U/L bit set to 0 = UAA
  - ↳ U/L bit set to 1 = LAA.

- In the context of IPv6 addresses/EUI-64, the meaning of U/L bit is reversed.
  - ↳ U/L bit set to 0 = MAC add. the EUI-64 int. was made from was LAA.
  - ↳ U/L bit set to 1 = The MAC address the EUI-64 int ID was from was a UAA

NOB: To learn more search eui-64 packetlife.net
- Eui-64 isn't a type of ipv6 add, its a method of automatically generating IPV6 add using specified prefix & MAC add.

Types of IPv6 address.

i) Global unicast
ii) Unique local
iii) Link local
iv) Anycast
v) Multicast
vi) Modified EUI-64

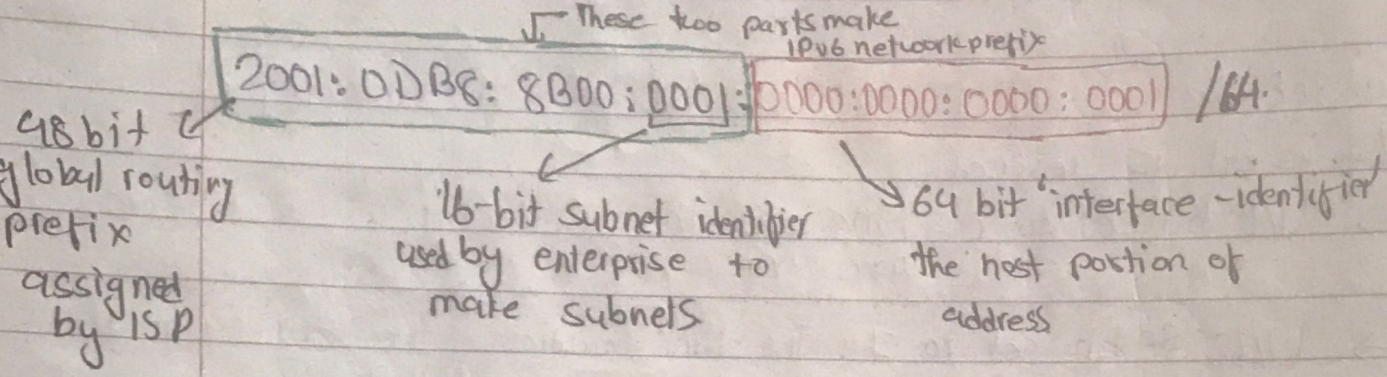
Global Unicast Address

- They are public addresses which can be used over internet
- Must register to use them. Because they are public addresses, it is expected that they are globally unique.
- Originally defined as 2000::/3 block (2000:: to 3FFF:FFF:.... FFFF)

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- Now defined as all addresses which aren't reserved for other purposes.

- Example of Global unicast address



Unique Local Address.

- These are like Private IPv4 addresses which can't be used over Internet.
- Unique local IPv6 add. are private add. & can't be be used over internet

- They can't be routed over internet.
- Unique We don't need to register to use them. They can be used freely within internal networks and don't need to be globally unique (*). ~~Can't be r~~
  (Though we need to try to make them unique)

- Uses the address block

  FC00::/7 (FC00:: to FDFF:FDFF:FDFF:....)

- However the later update requires the 8th bit to be set to 1, so first two digits must be FD.

FD | 45:93AC:8A8F | 0001:0000:0000:0000:0001 /64

↓ 40-bit 'global ID' which should be randomly generated.

* The global ID should be unique, so that addresses don't overlap when companies merge.

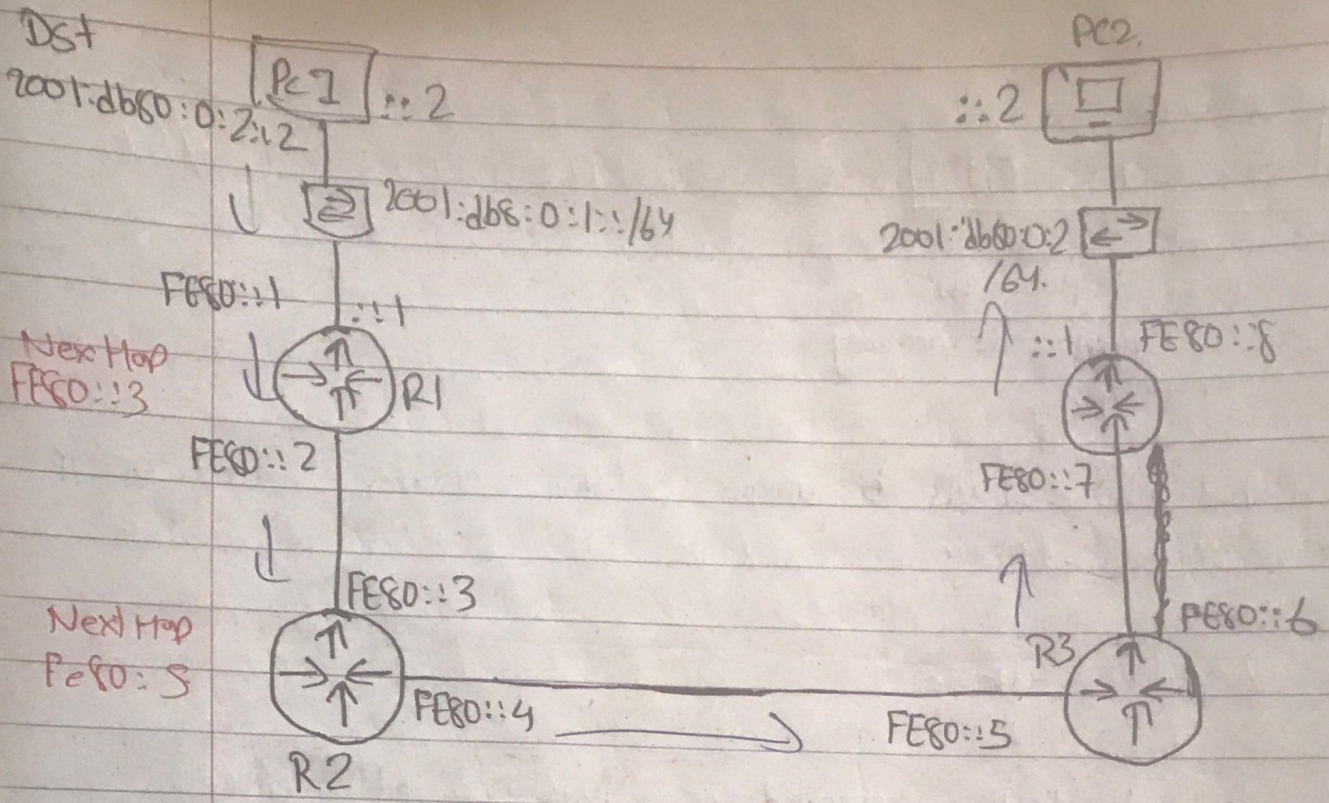
Link-Local addresses.

- They are automatically generated on IPv6-enabled interfaces.
- Use command R1(config-if)# ipv6 enable to enable IPv6 on an int, without actually configuring IPv6 add, on it.

- Uses the add. block FE80::/10 (FE80:: to FEBF:FFFF:----)
- However, the standard states that 54 bits after 5E80/10 should all be 0, so we won't see link local addresses beginning with FE9, FEA or FEB. Only FE8.

- The 'interface-id is generated using EUI-64.
- Link Local means that these add. are used for communication within a single link (subnet). Routers won't route packets with a link-local destination IPv6 add.

- Common uses of link-local addresses
  - ↳ routing protocol peerings (OSPFv3 uses link-local add. for neighbor adjacencies)
  - ↳ next-hop add. for static routes.
  - ↳ Next Neighbor Discovery Protocol (NDP), IPv6's replacement for ARP) uses link-local add. function.



Here if R1 tries to ping R3 it can't as if link local add & can't be routed. But it works fine for a next-hop

Multicast address

↳ Unicast addresses are one-to-one
One source to one destination

- Broadcast addresses are one-to-all
One source to all destinations (within the subnet).

- Multicast addresses are one-to-many.
One source to multiple destinations (that have joined specific multicast group).

- IPv6 uses range FF00::/8 for multicast (FF00:: to ffff:...)

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- IPv6 doesn't use broadcast (there is no 'broadcast add' in IPv6)

Purpose	IPv6 address	IPv4 address
All nodes/hosts (fxn like broadcast)	FF02::1	224.0.0.1
All routers	FF02::2	224.0.0.2
All OSPF routers	FF02::5	224.0.0.5
All OSPF DRs/BDRs	FF02::6	224.0.0.6
All RIP routers	FF02::9	224.0.0.9
All EIGRP routers	FF02::A	224.0.0.10

These addresses uses the 'link local' scope, which stays in local subnet.

Multicast add scopes

*) IPv6 define multiple multicast 'scopes' which indicate how far the packet should be forwarded.

* IPv6 multicast scopes
* Interface Local
  The packet doesn't leave local device. Can be used to send traffic to a service within local device.

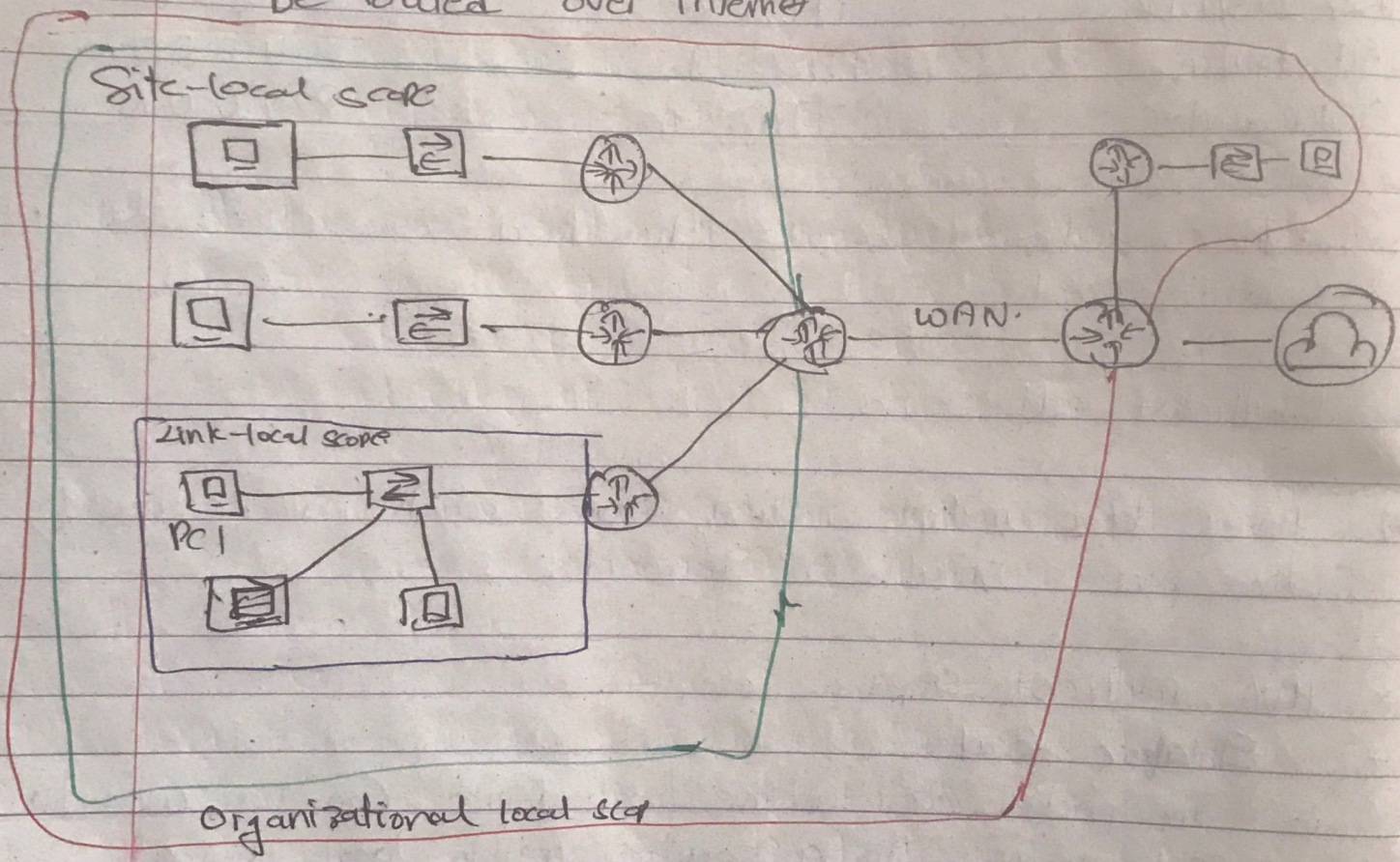
* Link-Local (FF02)
  The packet remain in local subnet. Routers won't route packet between subnets.

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* Site-local (FF05): The packet can be forwarded by routers. Should be limited to a single physical location (not forwarded over a WAN)

* Organization-local (FF08):
  Wider in scope than site-local (an entire company/org)

* Global (FF0E): No boundaries. Possible to be routed over internet



Global-Scope all this

Anycast address.

- Anycast is a new feature of IPV6.
- Anycast is 'one-to-one-of-many'.
- Multiple routers are configured with the same IPV6 address.
  - ↳ They use a routing protocol to advertise the address.
  - ↳ When hosts send packets to that destination add, routers will forward it to the nearest router configured with that IP add (based on routing metric)

- There's no specific add. range for anycast addresses. Use a regular unicast address (global unicast, unique local) & specify its as anycast address.

R1(config-if) # ipv6 address 2001:db8:1:1::99/128 any cast

* ~~IPv4~~
* Other IPv6 add.

* :: = The unspecified IPv6. add
  - Can be used when a device doesn't yet know its IPv6 add
  - IPv6 ~~add~~ default routes are configured to ::/0
  - IPv4 equivalent 0.0.0.0

* ::1 = The loopback address.
  - Used to test the protocol o stack on local device
  - Messages sent to this add. are processed within the local dev but not sent to other devices.
  - IPv4 equivalent : 127.0.0.0/8 add range.

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